

Truncated Marginal Neural Ratio Estimation (TMNRE) large A Simulation-Based Inference Approach

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Abstract

This report presents Truncated Marginal Neural Ratio Estimation (TMNRE) as a method for Bayesian inference when a simulator is available but the likelihood cannot be evaluated. It focuses on neural ratio estimation, low-dimensional marginal posteriors, and iterative truncation of the prior. Applications to the cosmic microwave background, strong gravitational lensing, and the sky-averaged 21-cm signal illustrate the capabilities and assumptions of TMNRE.

1 Introduction

Bayesian inference asks how an observation changes the plausibility of unknown parameters. With parameters $\boldsymbol{\theta}$ and observed data $\mathbf{x}_0$,

$$p(\boldsymbol{\theta}|\mathbf{x}_0) = \frac{p(\mathbf{x}_0|\boldsymbol{\theta})p(\boldsymbol{\theta})}{p(\mathbf{x}_0)}. \quad (1)$$

The prior describes plausible parameters before the observation, the likelihood describes how compatible the observation is with them, and the posterior is the updated distribution.

The difficulty addressed by TMNRE is that many scientific models are specified by a stochastic simulator. The simulator may generate $\mathbf{x}$ for a given $\boldsymbol{\theta}$, while the probability density $p(\mathbf{x}|\boldsymbol{\theta})$ may remain unavailable because the simulator includes latent physical processes, instrument effects, and noise. The original TMNRE paper presents the method for this combination of intractable likelihoods and high-dimensional parameters [1].

2 What simulation-based inference changes

Instead of evaluating a likelihood, training examples are generated:

$$\boldsymbol{\theta}^{(i)} \sim p(\boldsymbol{\theta}), \quad \mathbf{x}^{(i)} \sim p(\mathbf{x}|\boldsymbol{\theta}^{(i)}). \quad (2)$$

Each pair indicates what data can arise from a particular parameter choice. SBI learns from many such pairs.

A full posterior can still be difficult to learn. If the useful range is a fraction f of the prior range in each of d dimensions, the useful volume is approximately f^d . For $f = 0.1$ and $d = 20$, this is 10^{-20} . This is why TMNRE does not try to represent every detail of a high-dimensional joint posterior when the scientific result may require only a few one- and two-dimensional distributions.

3 Neural ratio estimation

TMNRE estimates the ratio

$$r(\mathbf{x}|\boldsymbol{\theta}) = \frac{p(\mathbf{x}|\boldsymbol{\theta})}{p(\mathbf{x})} = \frac{p(\boldsymbol{\theta}|\mathbf{x})}{p(\boldsymbol{\theta})}. \quad (3)$$

This ratio can be estimated with a classifier. Joint pairs are generated by drawing a parameter and simulating data from it:

$$(\boldsymbol{\theta}, \mathbf{x}) \sim p(\boldsymbol{\theta})p(\mathbf{x}|\boldsymbol{\theta}).$$

Independent pairs are made by combining independently drawn parameters and data:

$$(\boldsymbol{\theta}, \mathbf{x}) \sim p(\boldsymbol{\theta})p(\mathbf{x}).$$

The classifier learns whether the parameter and data are associated. If s is the probability assigned to the joint class and the classes are balanced, then

$$\frac{s(\mathbf{x}, \boldsymbol{\theta})}{1 - s(\mathbf{x}, \boldsymbol{\theta})} = \frac{p(\mathbf{x}|\boldsymbol{\theta})}{p(\mathbf{x})}. \quad (4)$$

Thus classification supplies a likelihood-to-evidence ratio without requiring an explicit likelihood expression.

4 Marginal neural ratio estimation

The final result does not usually require every model parameter. Writing $\boldsymbol{\theta} = (\boldsymbol{\phi}, \boldsymbol{\psi})$, where $\boldsymbol{\phi}$ are parameters of interest and $\boldsymbol{\psi}$ are nuisance parameters, the desired marginal is

$$p(\boldsymbol{\phi}|\mathbf{x}_0) = \int p(\boldsymbol{\phi}, \boldsymbol{\psi}|\mathbf{x}_0) d\boldsymbol{\psi}. \quad (5)$$

The nuisance parameters are not deleted from the model. They remain in the simulations and therefore affect the distribution of the simulated data. They are simply integrated out of the reported result. TMNRE estimates selected one- and two-dimensional marginals rather than a single very high-dimensional density.

This is a useful distinction: marginalisation is based on the scientific question, whereas truncation is based on the posterior support for the particular observation.

5 Truncating the prior

TMNRE proceeds in rounds. First, simulations are drawn from the original prior and used to train marginal ratio estimators. The estimators are evaluated on the observation to identify regions with appreciable posterior mass. A retained region Γ is then defined using

$$p_{\Gamma}(\boldsymbol{\theta}) = \frac{p(\boldsymbol{\theta})\mathbb{I}_{\Gamma}(\boldsymbol{\theta})}{\int_{\Gamma} p(\boldsymbol{\theta}) d\boldsymbol{\theta}}. \quad (6)$$

Within Γ , the relative prior probabilities are preserved; outside it, subsequent simulations assign zero mass. New simulations are therefore concentrated in a region more relevant to the observed data.

The truncation must be conservative. If an early estimate removes a genuine posterior mode, later rounds cannot recover it. Truncation should therefore be interpreted as an efficiency device rather than as a means of forcing the answer into a preferred range. The process stops when successive retained regions change very little.

6 Application 1: the CMB

Cole et al. apply TMNRE to measurements of the cosmic microwave background (CMB) [2]. The CMB contains information about cosmological parameters, but the analysis can also include many nuisance parameters. The paper reports converged one- and two-dimensional posteriors with orders of magnitude fewer simulator calls than conventional MCMC in its test problem. It also finds that the required simulation count is effectively independent of the number of nuisance parameters, because the method targets marginals rather than the full joint posterior.

Local amortisation is particularly important. The trained estimator can be evaluated on many nearby mock observations. This makes coverage and consistency tests practical. Since the true posterior is unknown, a credible interval cannot simply be trusted because it looks smooth. Testing whether nominal 68 percent regions contain the true value about 68 percent of the time is part of making the result credible. The application uses the open-source `swyft` implementation.

7 Application 2: strong gravitational lensing

A strong-lensing image contains a bright background source distorted by a foreground galaxy. Smaller dark-matter halos, including halos along the line of sight, perturb the image subtly. An individual perturbing halo may be hard to detect, while the population can still contain information about the warm-dark-matter mass function.

Anau Montel et al. combine parametric lens models with TMNRE and simulated sets of Hubble-resolution images to constrain the warm-dark-matter cutoff scale [3]. Their method marginalises over a large population of realistic perturbers as well as uncertainty in lens and source parameters. This matters because ignoring these objects or compressing the image into a hand-designed summary statistic can lose information or understate uncertainty.

Coogan et al. study the related problem of individual subhalo measurements [4]. Their mock-data analyses show that TMNRE can infer subhalo parameters while marginalising over hundreds of subhalos and line-of-sight halos. This provides a concrete example of why marginalisation is not merely a mathematical convenience: the nuisance population is physically real, and omitting it would make the model less realistic.

8 Application 3: the sky-averaged 21-cm signal

The sky-averaged 21-cm signal from the Cosmic Dawn and Epoch of Reionization can reveal information about early galaxies and cosmology. In practice, an analysis must account for foreground emission, beam uncertainties, ionospheric effects, and receiver noise. These effects make a simple likelihood model unreliable or computationally expensive.

Saxena et al. apply TMNRE to mock observations for the REACH experiment [5]. Their simulations include beam-weighted foregrounds, a physically motivated 21-cm signal, and radio-metric noise. They estimate posteriors for model parameters and find that combining multiple time slices and antennas improves constraints and exploration of the cosmological signal.

This example clarifies marginalisation. The target parameters might describe the astrophysical 21-cm signal, while foreground and instrument parameters are still simulated but treated as nuisance quantities. Truncation then directs later simulations towards combinations of signal and nuisance settings that can plausibly reproduce the observed spectrum. These nuisance settings must remain in the simulator; otherwise the final uncertainty would be unrealistically narrow.

9 What TMNRE achieves

Across these applications, three linked advantages emerge. First, the ratio formulation avoids direct likelihood evaluation. Second, marginal targets avoid requiring a detailed model of every parameter dependence. Third, truncation prevents later simulation rounds from spending most of their effort in regions already shown to be incompatible with the observation.

The CMB example demonstrates efficiency and local amortisation. Strong lensing demonstrates the ability to retain complicated populations of nuisance objects. The 21-cm example demonstrates that foreground and instrument effects can be included in a physically motivated simulator without requiring a simple analytic likelihood.

10 Limitations and questions

TMNRE does not remove the need for modelling choices. The prior determines where the initial simulations are drawn from, and the simulator determines which forms of uncertainty can appear in the training data. A misspecified simulator can produce a precise but misleading posterior.

Truncation introduces an irreversible risk: an early mistake can discard a later-important region. Neural classifiers can also be imperfect, particularly in narrow or multimodal problems. Consequently, posterior plots constitute incomplete evidence without coverage tests, simulation-based calibration, posterior-predictive checks, and comparisons with a trusted method where one is available.

The method also produces selected marginals rather than automatically giving every dependence in the full parameter space. This is a strength when the scientific question is focused, but a limitation if later questions require a marginal that was never trained.

11 Conclusion

TMNRE provides a practical compromise between physical realism and computational cost. The method accommodates detailed stochastic simulators, learns parameter–data associations through classification, focuses on low-dimensional marginals relevant to the scientific question, and iteratively concentrates simulations around the observed data.

The five applications show that this is not limited to one type of dataset: it has been used for CMB measurements, strong-lensing images, and 21-cm radio observations. Its credibility depends on realistic simulations, conservative truncation, explicit treatment of nuisance parameters, and empirical tests of calibration.

References

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